# Ibrahim Alam

(318) 617-4665 | ibalam@cs.unc.edu| LinkedIn | Professional Website

## Education

**University of North Carolina at Chapel Hill** **Expected May 2028**
**Master of Science in Computer Science (MSCS)**
*Focus: Machine Learning, Active Learning, Human-Computer Interaction*
**Louisiana State University (LSU), Baton Rouge, Louisiana** **Received May 2026**
**BS, Computer Science - Software Engineering | Minor: Mathematics** **GPA: 4.267**
*Roger Hadfield Ogden Honors College*
**Awards and Honors:** LSU Stamps Scholarship Recipient, LSU Distinguished Researcher, Tau Beta Pi Engineering Honors Society Member, LSU CXC Distinguished Communicator, LSU Tiger Twelve Member, McLaughlin Medal (highest GPA in College of Engineering), University Medal, College Honors, Summa Cum Laude

## Technical Skills

**AI/ML:** Python, PyTorch, HuggingFace, Weights & Biases, TensorFlow, scikit-learn, NumPy, pandas, distributed training, RAG architecture, agent systems, vLLM, pgvector, prompt engineering
**Software Development:** Java, C#, C++, JavaScript, SQL, Django, ASP.NET MVC/Core, ReactJS, PostgreSQL, REST APIs
**Infrastructure & Tools:** Azure, Git, Docker, Cursor, Claude Code, Codex, Microsoft TFS

## Experience

**Graduate Researcher | VisuaLab, UNC Chapel Hill | Advisor: Dr. Danielle Szafir | Aug 2026 – Present**

- Designing a clinician-in-the-loop active learning system for identifying precursor signs of freezing of gait in Parkinson's disease, extending a CHI 2024 visual analytics platform for clinical motion data
- Investigating how interface presentation order, model prediction first vs. raw data first, shapes expert labeling behavior, downstream model calibration, and overreliance on incorrect predictions

**Research Assistant | LSU Division of Computer Science and Engineering, Louisiana State University, Baton Rouge | Advisor: Dr. James Ghawaly | AISX Lab | Project - PECAN (Programming Encoder Classification Analysis Network) | (September 2024 - present)**

- Trained encoder models on 28M+ samples (319 languages), achieving 95.8% accuracy
- Benchmarked fine-tuned encoders against GPT-OSS-120B, Linguist, highlight.js, and Guesslang, including a Guesslang baseline retrained on our dataset to isolate architecture gains from data gains
- Built scalable training and evaluation pipelines with distributed training and hyperparameter tuning
- Conducted error analysis to identify model weaknesses and improve robustness
- Co-authoring a research paper for submission to FORGE (Conference on AI Foundation Models and Software Engineering)

**Machine Learning Software Developer Intern | LSU Office of Academic Affairs | Project – MikeGPT | (May 2025 – May 2026)**

- Engineered a RAG system over 38,000+ documents, improving answer accuracy and relevance for 30,000+ users
- Built backend services using Python, Django, PostgreSQL, and Azure with a staging service for controlled deployment
- Designed and implemented file encryption and secure document storage pipeline for sensitive institutional data
- Built a custom agent engine for complex multi-step queries, enabling context-aware response generation
- Conducted retrieval optimization and prompt engineering, significantly improving response quality and user satisfaction

**Software Developer | Tri-Core Technologies LLC, Baton Rouge | (May 2024 – present)**

- Evaluating and improving an internal RAG system for Red Stick Solutions, a cybersecurity and AI-governance firm
- Designing an evaluation framework to measure retrieval quality and answer accuracy
- Maintained and improved .NET full-stack systems for the Louisiana Department of Insurance (LDI), reducing downtime
- Delivered 10+ features and bug fixes with senior engineers, improving performance and efficiency
- Optimized producer/adjuster search in the Regulatory Management System, reducing lookup times for regulatory data
- Enhanced RMS and Employee Portal for hundreds of employees and thousands of users at LDI

## Presentations

**Alam I.** PECAN: Programming Encoder Classification Analysis Network. LSU Roger Hadfield Ogden Honors College Undergraduate Thesis Defense. May 2026; Louisiana State University, Baton Rouge, Louisiana.
**Alam I**. PECAN: *Programming Encoder Classification Analysis Network.* Stanford Research Conference April 2026; Stanford University, Stanford, CA.
**Alam I,** Biral B, Nguyen J. *MikeGPT: Enhancing LSU with AI*. LSU Discover Day Undergraduate Research and Creativity Conference. 2025 April 25; Louisiana State University, Baton Rouge, Louisiana